

A Bayesian Detection of Epistasis
Using Summarized Synergy Networks
in Islet Autoimmunity



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Dedication

To Andrew Gavin Olguin, this is for you.

I could not have done it without you.

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Abstract

Type 1 Diabetes (T1D) is a complex autoimmune disease characterized by the erroneous destruction of islet β -cells resulting in life-long dependence on exogenous insulin. Although T1D is considered to be largely genetic, with the heritability of the disease around 80%, roughly 20% of the phenotypic variance is still unexplained. Epistasis, the interaction of genetic variants, is thought to contribute to this missing heritability in T1D, yet the epistatic landscape of T1D and its precursor Islet Autoimmunity (IA) remains understudied. A genome-wide analysis of the epistatic architecture of IA could inform potential biological pathways in the etiology of T1D in addition to elucidating aspects of its missing heritability.

After quality control, 359 DAISY children selected for nested case-control study with a customized Exome array imputed to the TOPMed r2 reference panel were included for analysis of genome-wide epistasis detection. Within-gene synergy networks were used to capture non-linear epistatic effects and summarized through diffusion kernel principal components (kPC). Between-gene epistasis was detected through a bayesian interaction analysis of kPC-summarized genes.

RESULTS

DISCUSSION

1 Introduction

1.1 Type 1 Diabetes & Islet Autoimmunity

Type 1 Diabetes (T1D) is a complex autoimmune disorder where the immune system erroneously targets and destroys insulin-producing pancreatic islet- β cells, leading to a lack of insulin and elevated blood glucose levels²⁰. Unlike *monogenic* diseases – like sickle cell disease or cystic fibrosis, which result from variation within a single gene and are relatively rare – *complex* diseases are influenced by multiple chromosomal regions containing genetic variation, known as loci¹. The pathogenesis of T1D is believed to follow a continuous progression, beginning with the appearance of islet autoantibodies, followed by β -cell destruction, the onset of dysglycemia, and ultimately, the manifestation of hyperglycemia-related symptoms. Its preclinical stage, known as Islet Autoimmunity (IA), is characterized by the presence of autoantibodies prior to clinical diagnosis^{20,28}. However, the precise onset of IA remains somewhat ambiguous. The definition used by the Diabetes AutoImmunity Study in the Young (DAISY) cohort classifies IA as the presence of one or more islet autoantibodies on two or more consecutive visits, or a single autoantibody-positive visit followed by a Type 1 Diabetes diagnosis at the next visit¹⁴.

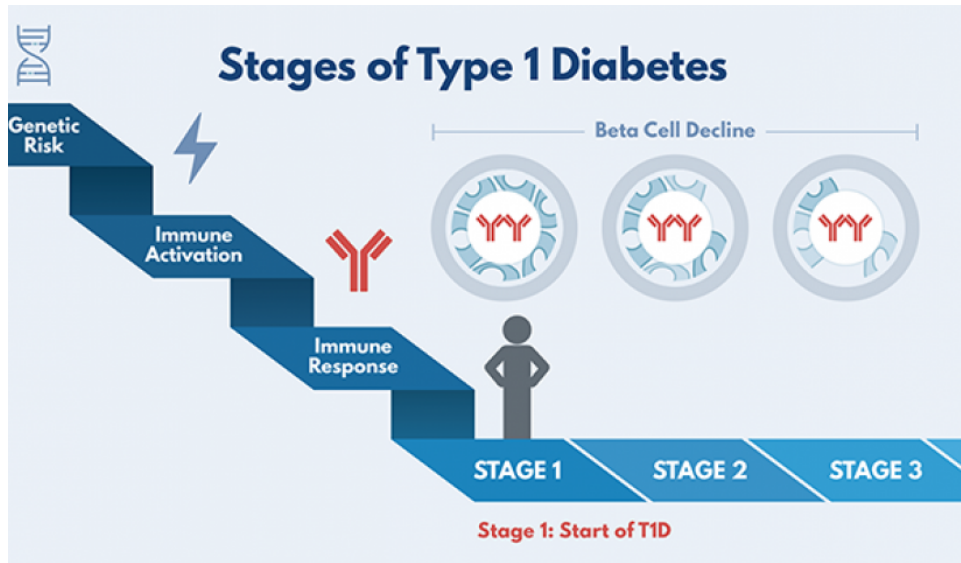


Figure 1: **Stages of T1D**

The progression of T1D follows three recognized stages. **Stage 1** marks the onset of Islet Autoimmunity (IA), with the presence of two or more diabetes-related autoantibodies but normal blood glucose levels. **Stage 2** is characterized by worsening β -cell function and emerging dysglycemia, despite the absence of symptoms. **Stage 3** is the clinical onset of T1D, marked by symptomatic hyperglycemia requiring insulin therapy^{2,28}.

Often diagnosed in early childhood, T1D left untreated can result in serious complications such as diabetic ketoacidosis, a life-threatening condition. With the disease incidence rising⁵, efforts to understand

the pathogenesis and etiology of T1D continue with the hopes of phase-specific therapeutic intervention to mitigate life-threatening complications, eliminate the burden of insulin pumps, and improve quality of life.

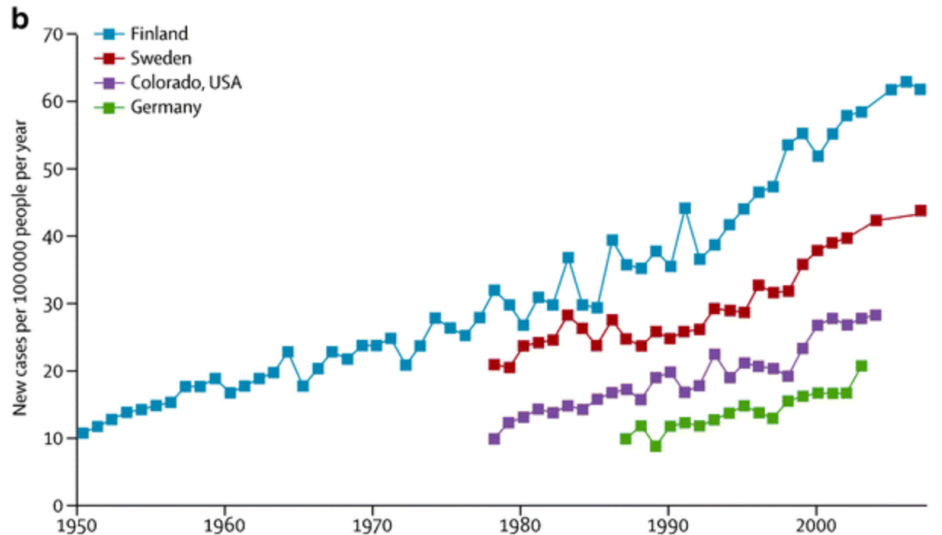


Figure 2: **Incidence of T1D in children aged 0-14 years**

The incidence of T1D has steadily increased across multiple countries over the past several decades, as shown in this longitudinal dataset⁵.

T1D is primarily considered a genetic disease, with environmental factors—such as viral infections—potentially influencing disease progression and the onset of IA. Ongoing research continues to investigate the viral onset hypothesis and the role of the complement system, which is primarily encoded within the Human Leukocyte Antigen (HLA) region²⁸. However, the focus of this research remains on genetic susceptibility.

The HLA region represents the strongest and longest-known genetic risk factor for T1D. The first association between HLA and juvenile-onset T1D was reported in the early 1970s through serological HLA typing³⁸. This finding was later validated in the 1980s using targeted genetic association testing, which pinpointed specific HLA-DQB1 alleles as key determinants of T1D susceptibility⁴⁰. This highly polymorphic region has been associated with many other autoimmune diseases and constitutes roughly 40% of the heritability of T1D²⁸.

Despite decades of research, questions about the genetic architecture of T1D remain. Over 60 candidate loci, collectively explaining roughly 80% of the heritability²⁹, have been identified with respect to T1D. However, this leaves approximately 20% of heritability unexplained, a phenomenon known as the missing heritability problem. Addressing this gap requires further investigation into rare variants, common variants with small effect sizes, and gene-gene interactions (epistasis)^{29,33}.

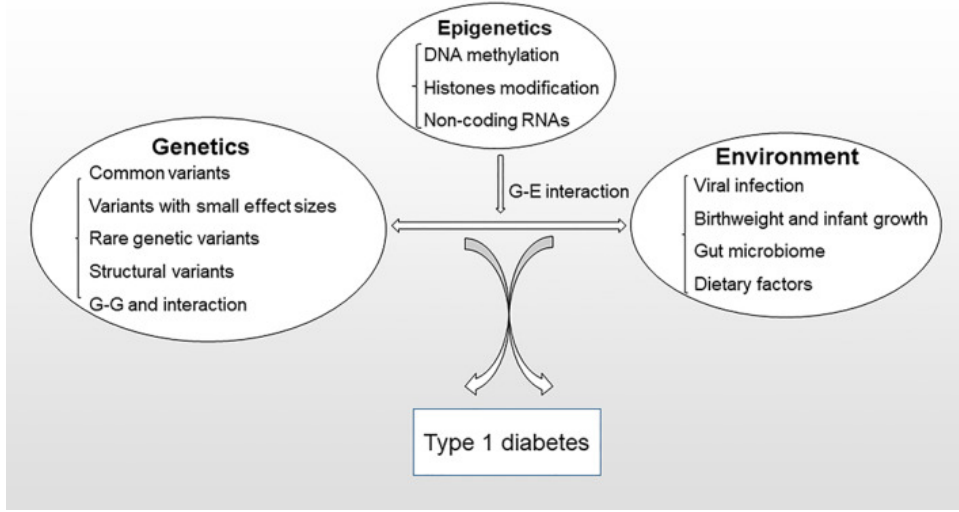


Figure 3: **Potential sources of the missing heritability in T1D**

Here G-G interaction is a gene by gene interaction and G-E is gene by environment interaction²⁹.

1.2 Epistasis

Epistasis has been defined and redefined many times since its coinage by Bateson in 1909, often leading to ambiguity in its definition. Seldom mentioned, however, is that Bateson also defined the term *hypostasis*^{6,10}. These two terms were initially meant to describe a hierarchical relationship where a higher-level factor (epistatic gene/allele) controls whether a lower-level factor (hypostatic gene/allele) can be expressed. Here, *epistatic* and *hypostatic* derive from the Greek prefixes *epi* meaning "upon", *hypo* meaning "under", and root *stasis* meaning "standing". Thus, to Bateson, an epistatic gene "stands upon" a hypostatic gene, masking the hypostatic gene's expression.

Epistasis, or *dual epistacy*, was redefined under a statistical context by Fisher in 1918 as a deviation from linearity in allelic effects with respect to some quantitative phenotype $D^{10,13}$. Here, we already see differing definitions of epistasis emerging just 9 years apart, each carrying distinct implications. While Fisherian statistical epistasis does not necessarily imply biological or functional epistasis, it provides a framework for detecting genetic interactions that warrant further investigation. The modern definition of epistasis is largely an umbrella term for genetic interaction. Under this umbrella, we find the term *positive/negative epistasis*, which refers to the interaction between genetic variants where positive (negative) epistasis occurs when their combined effect on a trait or disease risk is greater (less) than the sum of their individual effects. At the population level, this deviation from additivity plays a crucial role in shaping genetic architecture and disease susceptibility¹⁹.

Genome-wide association studies (GWAS) are a modern approach to discover genetic associations with a disease. GWAS test millions of representative single nucleotide polymorphisms (SNPs) from a

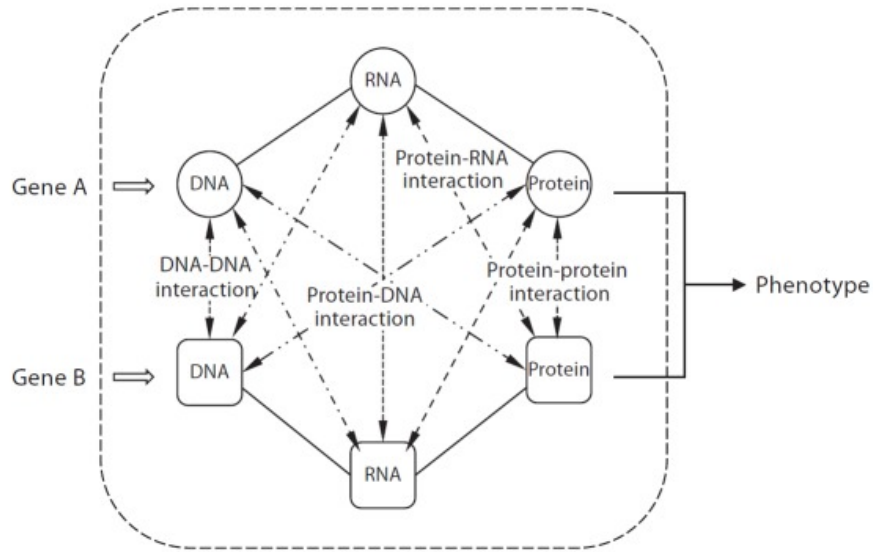


Figure 4: Gene-Gene Interaction

This is a conceptual model showing how Gene-Gene interactions play a role in phenotypic variation through multiple molecular interactions, including DNA-DNA, protein-DNA, protein-RNA, and protein-protein interactions⁴².

population in a linear framework where each SNP is tested for association with a phenotype individually. The standard correction to account for multiple hypothesis testing in GWAS is the Bonferroni correction, which adjusts for the number of independent SNPs in the genome. This method typically applies a genome-wide significance threshold of 5×10^{-8} for European populations to identify statistically significant associations and similar but more stringent threshold for African populations^{11,35}.

There are issues with this standard of practice. The standard genome-wide significance threshold faces challenges in evolving GWAS practices. It was developed for common variants, potentially lacking power for rare variants. It does not take into account the prior probability of a variant being associated with a phenotype, nor does it account for the statistical power of the test which both influence the interpretation of the p -value. These oversights can distort results, particularly when considering epistasis, where interactions between variants may not align with assumptions of independence and testing SNPs in pairs or tuples of length n drastically increase the already large number of hypotheses. Applying the standard threshold universally risks missing meaningful associations while inflating false positives, especially in diverse populations or complex phenotypes^{18,35}.

Genome-wide association interaction (GWAi) studies are an extension of GWAS aimed at epistasis detection^{8,16}. There are two categories of methods that emerge when considering GWAi: exhaustive and

non-exhaustive¹⁶. Non-exhaustive methods avoid the challenge of multiple testing correction by limiting the analysis in some way. Typically, these analyses determine candidate genes a priori to test, use GWAS results to look at interactions between significant SNPs overlooking those SNPs with small effects with potentially large interaction effects, or use some sort of dimensionality reduction^{3,16}. To address the burden of multiple testing in exhaustive approaches, Bayesian methods have been recommended for both GWAS and GWAI^{18,24,35,39,43}.

1.3 Bayesian Statistics

Bayesian statistics is named after Reverend Thomas Bayes, a part-time mathematician who was among the first recorded individuals to attempt solving the problem of inverse probability^{15,37}. Inverse probability seeks to determine the probability of a cause given an observed outcome as opposed to classical probability which estimates the likelihood of an event given known conditions. In Bayes' work that was posthumously communicated by his friend and colleague Richard Price, he posed the problem

"*Given* the number of times in which an unknown event has happened and failed: *Required* the chance that the probability of its happening in a single trial lies somewhere between any two degrees of probability that can be named."⁷

Bayes' original work did not explicitly introduce what we now know as Bayes' theorem:

$$p(\theta | y) = \frac{p(y | \theta)p(\theta)}{p(y)}$$

This was instead formulated by Laplace. Bayes' contribution was rather the concept of compromising prior information, $p(\theta)$, with information from observed data, $p(y | \theta)$. He introduces this concept with an analogy to a billiard table where he considers a scenario where a ball is randomly thrown onto a unit interval, and the goal is to estimate the probability of a future event occurring based on where the ball landed. Bayes introduced what we would now call a uniform prior distribution on θ such that θ was modeled as the probability parameter of a binomial distribution^{7,12}.

In population genetics, the classical frequentist approach to discovering associations with a phenotype typically involves computing the p -value for the null hypothesis (H_0) of no association. Despite our desire to interpret the p -value as the probability of H_0 being true (i.e. $p\text{-value} < 0.05$ implying significance through rejecting the null in favor of the alternative hypothesis H_1), this is simply not the case. The p -value relies not only on the how unlikely it is under H_0 but also how unlikely it is under H_1 which differs from study to study due to differences in power and measures the extremeness of the result

under the null²². A fundamental limitation of the p -value aside from its comparability across studies is that with-in a study, two genetic associations with equivalent p -values may have different levels of true association due to underlying factors such as minor allele frequency (MAF) which is the frequency of the non-dominant allele in the reference population. Some of these limitations can be addressed using the Bayesian framework, although doing so comes at the cost of increased modeling assumptions³⁹.

Modern Bayesian approaches incorporate prior probabilities and probabilistic modeling to improve statistical power and control false positives. In the context of a frequentist GWAS, the number of comparisons needs to be corrected for. The Bayesian perspective avoids this requirement as the p -value analogue is the *posterior probability*, $P(H_0 = \text{TRUE} \mid \text{Data})$, and is truly interpreted as a probability of true association conditioned on the observed data²².

1.4 Motivation

Epistasis is particularly important in complex autoimmune diseases, where interactions between immune-related loci can shape immune system function and the loss of self-tolerance³³. T1D is a prime example of this. While significant progress has been made in identifying genetic risk loci, including the well-established role of the HLA region, a substantial proportion of heritability remains unexplained²⁸. This missing heritability may be partially attributed to epistasis, which is not captured by standard GWAS²⁹.

Traditional GWAS frameworks are optimized for detecting marginal effects of individual SNPs but are not well-suited for modeling the joint effects of interacting loci. When considering epistasis, the number of SNP pairs (or n -tuples) to test increases combinatorially, exacerbating the multiple testing problem and further reducing power. Furthermore, p -values in the frequentist framework are often misinterpreted, and their interpretation can vary substantially depending on study-specific factors^{22,41}.

Bayesian methods offer a compelling alternative. They are not only well-suited for discovery in large-scale genomic studies but also allow for probabilistic modeling of uncertainty and incorporation of prior knowledge. Unlike p -values, posterior probabilities are directly interpretable and provide a more intuitive framework for assessing evidence of association. In the context of epistasis, the Bayesian approach provides a natural way to mitigate the multiple testing problem while identifying subtle, biologically meaningful interactions that may contribute to disease risk⁴¹.

GWAI studies have been relatively understudied in T1D and the methods for epistasis detection continue to evolve along with computational ability. We extend the work of *Walakira et al.*⁴¹ and apply this extended method to IA, with the goal of elucidating the epistatic landscape that may shape the early progression of T1D.

2 Methodology

2.1 Overview

Within-gene synergy networks were used to capture non-linear epistatic effects in the DAISY cohort and summarized through diffusion kernel principal components (kPCs). Between-gene epistasis was detected through a Bayesian interaction analysis of kPC-summarized genes. Gene pairs with evidence of interaction were then included pairwise in an intelligible Random Forest known as Explainable Boost Machine (EBM) at the SNP level to identify SNPs contributing to IA interacting across genes.

2.2 Data Preparation

After quality control, 359 DAISY children selected for nested case-control study with a customized Exome array imputed to the TOPMed r2 reference panel were included for analysis of genome-wide epistasis detection.

	Control (N=187)	IA (N=172)	Overall (N=359)
Sex			
Male	102 (54.5%)	90 (52.3%)	192 (53.5%)
Female	85 (45.5%)	82 (47.7%)	167 (46.5%)
Non-Hispanic White			
No	31 (16.6%)	28 (16.3%)	59 (16.4%)
Yes	156 (83.4%)	144 (83.7%)	300 (83.6%)
DR3/4 Membership			
No	154 (82.4%)	112 (65.1%)	266 (74.1%)
Yes	33 (17.6%)	60 (34.9%)	93 (25.9%)

Table 1: **DAISY demographics included in genome-wide epistasis analysis**

Summary of sex, ethnicity (Non-Hispanic White), and HLA-DR3/DR4 genotype status, stratified by islet autoimmunity (IA) case-control status. DR3/4 refers to the high-risk HLA-DR3/DR4 haplotype associated with type 1 diabetes.

2.3 Summarizing Within-Gene Epistasis

To model complex *within-gene* SNP interactions that contribute to IA in DAISY children, we build from foundational concepts in information theory, such as entropy and mutual information, to quantify

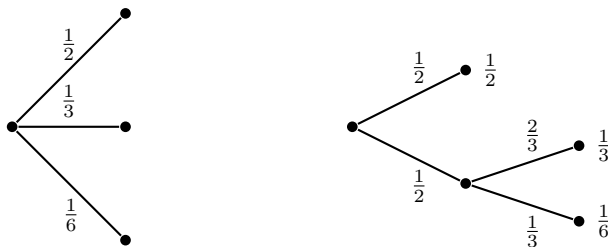


Figure 5: **Shannon's 3rd Property of H**

This is a decomposition of a choice from three possibilities. The **left tree** has *one choice* with probabilities $p_{l1} = \frac{1}{2}, p_{l2} = \frac{1}{3}, p_{l3} = \frac{1}{6}$ and the **right tree** has *two choices* with the first having equal probability $p_{r1} = p_{r2} = \frac{1}{2}$ and the second having probabilities $p_{r21} = \frac{2}{3}, p_{r22} = \frac{1}{3}$ ³⁶.

the epistatic contribution of SNP combinations. Using these quantities, we define pairwise synergy between SNPs and construct gene-specific synergy matrices reflecting potential non-linear effects. We then leverage the graph-theoretic properties associated with the Laplacian to model information flow across SNP-level interaction networks using diffusion kernels. Finally, by applying diffusion kernel principal component analysis (kPCA), we summarize each gene's within-gene epistatic architecture as a single, non-linear gene-level feature for downstream between-gene epistasis analysis.

2.3.1 Entropy

Claude Shannon, the father of information theory³⁴, introduced the concept of *entropy* in *A Mathematical Theory of Communication*³⁶ in 1948. Shannon proposed a measure of uncertainty $H(p_1, p_2, \dots, p_n)$ for probabilities p_i such that $i \in \{1, 2, \dots, n\}$ with the following desirable properties:

1. H should be continuous in the p_i .
2. If all the p_i are equal, $p_i = \frac{1}{n}$, then H should be a monotonic increasing function of n . With equally likely events there is more choice, or uncertainty, when there are more possible events.

$$H(\frac{1}{2}, \frac{1}{2}) < H(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}) < \dots < H(\frac{1}{n}, \dots, \frac{1}{n})$$

3. If a choice be broken down into two successive choices, the original H should be the weighted sum of the individual values of H .

$$H(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}) = H(\frac{1}{2}, \frac{1}{2}) + \frac{1}{2}H(\frac{2}{3}, \frac{1}{3})$$

Shannon proved (in Appendix 2 of *A Mathematical Theory of Communication*³⁶) that the only H satisfying these three properties is:

$$H = -K \sum_{i=1}^n p_i \log_b(p_i)$$

where b is the base of the logarithm determining the units ($b = 2 \implies \text{bits}$ and $b = e \implies \text{nats}$) and K is a positive constant equivalent to the choice of a unit of measure (i.e. if you choose $b = 2$ your units would be in *bits* but if you also choose $K = \frac{1}{\log_2(e)}$ you can convert it *nats*). For convenience, we will assume $b = 2$ and $K = 1$ and use the equivalent statement

$$H = \sum_{i=1}^n p_i \log\left(\frac{1}{p_i}\right) \quad (1)$$

for the remainder of the paper.

While Shannon's derivation is a rigorous result of three desirable properties, a more intuitive approach to constructing H starts with *surprise*^{27,34}.

Intuitive Justification. Let A be an event—for example winning the lottery. Let the probability of winning be $P(A) = p$. If p is small, the surprise associated with A , denoted $S(A)$, should be large. Let's assume $S(A) = \frac{1}{p}$ as it captures this idea and alter it as needed.

If $p = 1$, then $S(A) = 1$ but it would make more sense for the surprise to be 0. Now let $S(A) = \log(\frac{1}{p})$. If $p = 1$, then $S(A) = 0$. As $p \rightarrow 0$, $S(A) \rightarrow \infty$. This is exactly what we would want as a surprise function. Since we know $P(A) = p$ is the probability of winning the lottery, the probability of losing the lottery is $P(A^c) = 1 - P(A) = 1 - p$. The expected surprise associated with winning the lottery would then be

$$E[S(A)] = P(A) \cdot S(A) + P(A^c) \cdot S(A^c) = p \cdot \log\left(\frac{1}{p}\right) + (1 - p) \cdot \log\left(\frac{1}{1 - p}\right)$$

. If we let $p_1 = p$ and $p_2 = 1 - p$ then we can write:

$$E[S(A)] = \sum_{i=1}^2 p_i \log\left(\frac{1}{p_i}\right) = H.$$

Thus, entropy can be interpreted as the average surprise or the expected uncertainty of an outcome. $\square$

The figure below illustrates the entropy $H(p)$, or expected uncertainty, as a function of the probability p of an event A . Note the maximum entropy occurs when $p = 0.5$. As $p \rightarrow 0$ and $p \rightarrow 1$ the event is more predictable and thus the entropy approaches 0.

A natural extension is to consider the joint entropy of multiple events $X \in \mathcal{X}$, $Y \in \mathcal{Y}$ where $p(X, Y)$

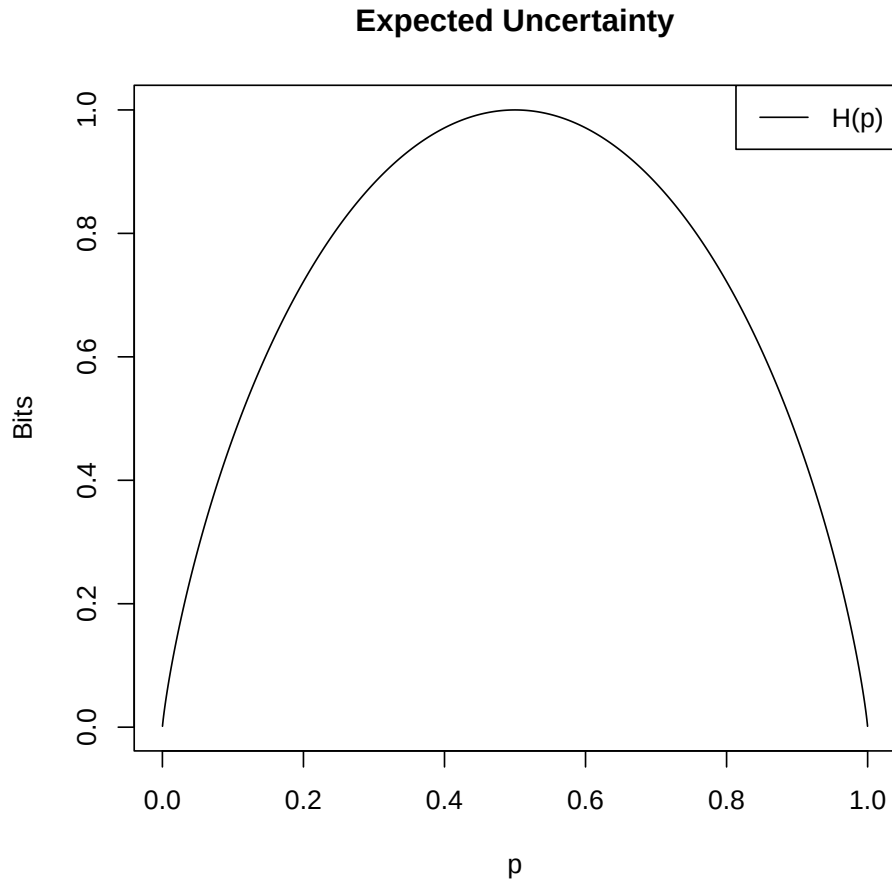


Figure 6: **Entropy as Expected Uncertainty**

Entropy $H(p)$, or the expected uncertainty of a binary outcome, is plotted as a function of the event probability p . The curve is maximized at $p = 0.5$, where uncertainty is highest. As $p \rightarrow 0$ or $p \rightarrow 1$, the outcome becomes more predictable and the entropy approaches 0^{34,36}.

is the joint probability. The entropy is then

$$H(X, Y) = \sum_{x,y} p(X, Y) \log\left(\frac{1}{p(X, Y)}\right) \quad (2)$$

where $H(X, Y) \leq H(X) + H(Y)$ with equality occurring if the event X and Y are independent such that $p(X, Y) = p(X)p(Y)$ ³⁶. Another natural extension is then to define the conditional entropy of X given Y :

$$H(X | Y) = \sum_{x,y} p(X, Y) \log\left(\frac{1}{p(X | Y)}\right) = H(X, Y) - H(Y) \quad (3)$$

$H(X, Y)$	
$H(X)$	$H(X Y)$
$H(Y X)$	$H(Y)$

Figure 7: **Entropy and Conditional Entropy**

The relationship between **Joint Entropy**, **Entropy** or Marginal Entropy, and **Conditional Entropy**²⁶.

2.3.2 Mutual Information

We now have a measure of the expected uncertainty associated with an event—namely, entropy—as well as a means of quantifying uncertainty in the presence of a dependent event, via conditional entropy. Figure 7 highlights the overlap in information being communicated from both events when they depend on one another. We can quantify this redundancy as *mutual information*, denoted:

$$I(X; Y) \equiv H(X) - H(X | Y) = H(Y) - H(Y | X) \equiv I(Y; X).$$

Note that $I(X; Y) \geq 0$ as $H(X | Y) \leq H(X)$. This measures the average reduction in uncertainty about X that results from having information about Y or can equivalently be thought of as the average information gain when knowing information about both events as highlighted in Figure 8^{17,26,41}.

$H(X, Y)$		
$H(X)$	$H(X Y)$	
$H(Y X)$	$H(Y)$	
$H(Y X)$	$I(X; Y)$	$H(X Y)$

Figure 8: **Entropy, Conditional Entropy, and Mutual Information**

The relationship between **Joint Entropy**, **Entropy** or Marginal Entropy, **Conditional Entropy**, and **Mutual Information**²⁶.

Importantly, the mutual information extends to joint variables. Let $X = (A, B)$. Then,

$$I(X; Y) = I(A, B; Y) = H(A, B) - H(A, B | Y) = H(Y) - H(Y | A, B) = I(Y; A, B) = I(Y; X).$$

The formulations of mutual information used throughout the remainder of this paper are:

$$I(A; D) = H(D) - H(D | A) \tag{4}$$

$$I(A, B; D) = H(D) - H(D | A, B) \tag{5}$$

where D denotes the *disease* outcome and A and B represent SNPs. Thus, the interpretation of $I(A, B; D)$ is the mutual information of the disease D and jointly knowing genotypes for SNP A and SNP B . This is equivalent then to the difference in the expected uncertainty in the disease D and jointly knowing genotypes for SNP A and SNP B . In other words, $I(A, B; D)$ reflects the reduction in uncertainty about the disease status provided by jointly knowing the genotypes for A and B ⁴¹.

2.3.3 Computing Within-Gene SNP Synergies

The concept of bivariate *synergism* quantifies the combined effect of SNPs S_i and S_j on a disease D beyond their individual contributions. It is calculated as:

$$Syn_{ij} \equiv Syn(S_i, S_j; D) = I(S_i, S_j; D) - [I(S_i; D) + I(S_j; D)] \quad (6)$$

A positive synergy, $Syn_{ij} > 0$, implies that the average reduction in the uncertainty about the disease from *jointly* knowing SNPs S_i and S_j is greater than the sum of their individual contributions. This indicates a deviation from linearity in their effects, suggesting a *positive epistatic* interaction influencing disease status.

Synergy can be extended to include not just pairwise interactions, but higher-order interactions. The general form for synergy with disease D and SNPs $S_1, S_2, \dots, S_n$ for $n \in \mathbb{N}$ where $\mathcal{S} = \{S_1, \dots, S_n\}$ is:

$$Syn(S_1, S_2, \dots, S_n; D) = \sum_{k=1}^n (-1)^k \sum_{\substack{s \subseteq \mathcal{S} \\ |s|=k}} I(s; D) \quad (7)$$

This formulation captures the deviation from additivity in the mutual information between the disease outcome and combinations of SNPs, accounting for double counting through the principle of inclusion and exclusion. However, the remainder of this paper will focus on bivariate synergy, where $n = 2$ as seen in Equation 6.

Using genotypes of SNPs annotated to the gene-level, we can construct pairwise synergies over all SNPs within each gene and define a *synergy matrix* $\mathbf{S}$, with $diag(\mathbf{S}) = 0$, where each off diagonal entry

$\mathbf{S}_{ij}$ corresponds to the pairwise synergy between SNPs i and j for $i \neq j$:

$$\mathbf{S} = \begin{pmatrix} 0 & \text{Syn}_{12} & \cdots & \text{Syn}_{1n} \\ \text{Syn}_{21} & 0 & \cdots & \text{Syn}_{2n} \\ \vdots & \cdots & \ddots & \vdots \\ \text{Syn}_{n1} & \text{Syn}_{n2} & \cdots & 0 \end{pmatrix}. \quad (8)$$

This synergy matrix enables the construction of SNP-level graphs within each gene, with interpretable edge weights reflecting potential pairwise epistatic interactions. To summarize the total pairwise epistatic contribution of each SNP, we define a diagonal matrix $\mathbf{D}$ whose entries are the row sums of $\mathbf{S}$:

$$\mathbf{D} = \begin{pmatrix} \sum_{j=1}^n \mathbf{S}_{1j} & 0 & \cdots & 0 \\ 0 & \sum_{j=1}^n \mathbf{S}_{2j} & \cdots & 0 \\ \vdots & \cdots & \ddots & \vdots \\ 0 & 0 & 0 & \sum_{j=1}^n \mathbf{S}_{nj} \end{pmatrix} \quad (9)$$

We now define a *Laplacian-esque* matrix $\mathbf{L}$ for each gene's synergy graph G that captures both the pairwise epistasis and the aggregate pairwise epistasis:

$$\mathbf{L}_{ij} = \begin{cases} -\text{Syn}_{ij} & \text{if } i \neq j, \\ \sum_{l=1}^n \text{Syn}_{il} & \text{if } i = j. \end{cases}$$

or alternatively

$$\mathbf{L} = \mathbf{D} - \mathbf{S} \quad (10)$$

2.3.4 Diffusion Kernel Principal Components as Gene Summaries

The Laplacian matrix has an intimate connection the *heat equation* presented in physics that describes how heat and other diffusing quantities spread through a medium. Diffusion kernels are often referred to as "heat" kernels due to it being the fundamental solution to the *heat equation*^{9,21}:

$$\frac{\partial u}{\partial t} = -\mathcal{L}u.$$

Applied to graphs, diffusion can be thought of as the spread of information from node to node. Imagine a graph or network where the edges and nodes are made of metal. If we apply enough heat to one node so that visibly see the metal turn red, we can watch as the heat transfers from the node to which it was applied to its neighbors. This captures the spread of information to its direct neighbors as well as its indirect neighbors as depicted in Figure 9. The *kernel* summarizes this diffusion of information between nodes²³.

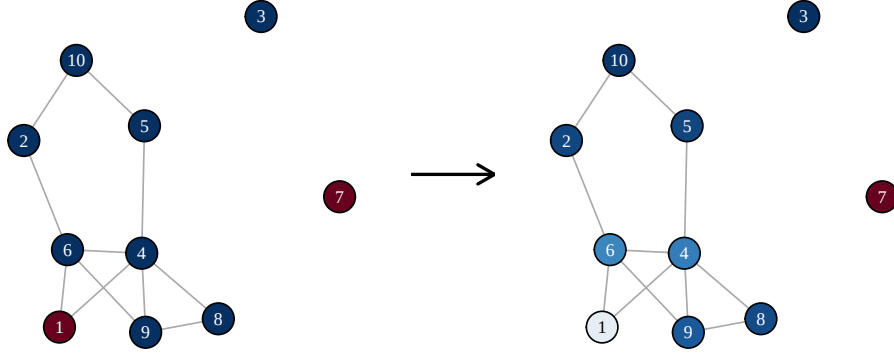


Figure 9: **Diffusion Intuition**

The graph on the **left** depicts nodes 1 and 7 as "hot", while the remaining nodes are "cold". After diffusion (graph on the **right**), node 1 has "cooled" as it transfers heat to its direct neighbors—and to a lesser extent, its indirect neighbors—demonstrating the spread of information. In contrast, node 7 remains "hot" due to its lack of connections.

To summarize each gene's epistatic landscape while capturing the non-linearity present, we apply *diffusion kernel principal component analysis* (diffusion kPCA) to $\mathbf{L}$. The *diffusion kernel* is defined as:

$$K_{L0} = e^{-\beta \mathbf{L}} \quad (11)$$

where $\beta \geq 0$ is a tuning parameter that controls the extent of diffusion or how easily information is able to be passed. As β increases, it is easier for information to spread between nodes. A β of 0 would imply no information is spread. Since the optimal value of β for modeling epistasis is unknown, we followed the approach in *Walakira et al.*⁴¹, allowing β to range from 0 to 10 in increments of 0.1 and averaging the resulting kernels to obtain the final kernel matrix:

$$K_L = \frac{1}{101} \sum_{i=1}^{101} K_{L0i} \quad (12)$$

K_L then summarizes the overall diffusion behavior, capturing how information spreads through the graph across a range of diffusion intensities, and in doing so, reflects the aggregate, pairwise, and higher-order epistatic architecture.

We can now apply this architecture to the genotype data to construct a kernel matrix K , which captures gene-level similarity between individuals based on each gene’s epistatic profile.

$$K = GK_L G^T \quad (13)$$

where G is an $n \times p$ genotype matrix for gene g , with n individuals and p SNPs. The resulting kernel matrix K is then centered and subjected to eigendecomposition using the RSpectra package³¹. The first principal component is then extracted and used as gene g ’s within-gene epistasis summary.

2.4 Between-Gene Epistasis Detection

To model complex *between-gene* interactions that contribute to IA in DAISY children, we transition from information-theoretic approaches to a *Bayesian semi-parametric*^{4,41} framework implemented in the NLIinteraction⁴ R package. This enables flexible modeling of non-linear gene-gene interactions while maintaining interpretability and mitigating the burden of hypothesis correction during discovery. After discovering which genes have evidence for pairwise interaction, we reintroduce the SNP-level data into an Explainable Boost Machine²⁵ (EBM) Random Forest to pinpoint specific variants driving cross-gene interactions contributing to IA.

However, before the explicit modeling of genes and gene-pairs, a summary measure for chromosomes was made using Gaussian Process regression so that a non-linear, phenotype informed summary measure could be formed. This summary measure was then subject to the same Bayesian Interaction model to discover and rank the importance of chromosomal pairs in relation to IA. This was conducted as a means to reduce the feature space. The Bayesian Interaction model at the gene level will only be run

on chromosomal pairs found to have evidence of inter-chromosomal interaction rather than exhaustively evaluating all possible gene-pairs across the genome.

2.4.1 Gaussian Process Regression

Gaussian Process regression (GPR) was used to generate a non-linear, phenotype-aware summary of gene variation for each chromosome. This approach enables the construction of a single, continuous latent feature per chromosome that captures complex genetic patterns associated with the phenotype, beyond what can be captured through linear dimensionality reduction methods such as PCA.

Let $X^{(c)}$ be the standardized genotype matrix for chromosome c , with individuals indexed by $i = 1, \dots, n$ and SNPs indexed by $j = 1, \dots, p_c$. The ancestry-corrected phenotype vector is denoted $Y \in \mathbb{R}^n$. GPR models the conditional distribution of Y given $X^{(c)}$ as a realization of a multivariate Gaussian:

$$Y \sim \mathcal{N}(0, K^{(c)}), \quad (14)$$

where $K^{(c)}$ is the $n \times n$ kernel matrix capturing pairwise similarity between individuals based on their genotype profiles on chromosome c . We use the Radial Basis Function (RBF) kernel, also known as the squared exponential kernel, defined as:

$$K_{ij}^{(c)} = \exp\left(-\frac{\|X_i^{(c)} - X_j^{(c)}\|^2}{2\sigma^2}\right) \quad (15)$$

where $X_i^{(c)}$ and $X_j^{(c)}$ are the genotype vectors of individuals i and j , and σ is a parameter controlling the smoothness of the kernel.

To extract a one-dimensional summary for each chromosome, we compute the posterior mean of the latent function $f^{(c)}$ under the GP model:

$$\hat{f}^{(c)} = K^{(c)}(K^{(c)} + \lambda I)^{-1}Y \quad (16)$$

where λ is a regularization parameter to account for noise. The resulting vector $\hat{f}^{(c)} \in \mathbb{R}^n$ represents a non-linear, phenotype-informed summary feature for chromosome c , which is then used as the input to the Bayesian interaction model described in Section 2.4.2.

This dimensionality reduction procedure allows us to retain biologically relevant signal while reducing the computational burden of modeling all gene-pairs directly, and only considering gene-pairs among chromosomal-pairs with evidence of interaction, rather than exhaustively evaluating all possible gene-pairs genome-wide.

2.4.2 Bayesian Interaction Modeling

For a pair of genes $\{g_1, g_2\} \subset \mathcal{G}$, where $\mathcal{G}$ is the set of all genes being analyzed, and for individual i , we take the first principal components from the within-gene epistasis summaries described in Section 2.3.4 and denote them individually as $\{X_{ig_1}, X_{ig_2}\}$ and their interaction as $X_{ig_1g_2}$. Interacting gene-pairs were determined to have evidence of epistasis based on a positive *posterior inclusion probability* (PIP) which quantifies the probability that a given feature—or pair of features—is included in the model conditioned on the observed data.

To flexibly model smooth, non-linear relationships (see Figure 10), we applied the natural spline basis function $ns(\cdot)$, implemented in R³², to each main effect X_g for all $g \in \mathcal{G}$, as well as to each interaction term $X_{g_1g_2}$ for all $\{g_1, g_2\} \subset \mathcal{G}$. We denote the $ns(\cdot)$ -transformed summaries as:

$$\tilde{X}_{g_1} = ns_1(X_{g_1}) \quad (17)$$

$$\tilde{X}_{g_2} = ns_2(X_{g_2}) \quad (18)$$

$$\tilde{X}_{g_1g_2} = ns_1(X_{g_1}) \cdot ns_2(X_{g_2}) \quad (19)$$

The Bayesian model formulation makes the assumption of a normally distributed response. However, the binary phenotype (IA vs. non-IA) was residualized using the first seven ancestry principal components, following the recommendation of Walakira et al.⁴¹. While this approach does not strictly satisfy the model assumptions, prior work³⁰ supports it as a computationally efficient approximation. Since our goal is not to perform direct inference, but rather to prioritize or rank genes and gene pairs for further investigation, this approximation is considered sufficient for discovery-oriented analysis.

Let Y be the ancestry corrected phenotype, p be the number of genes, i.e., $|\mathcal{G}| = p$, and let k be a number sufficiently large such that all gene effects can be captured by the model. We assume the corrected phenotype is drawn from a normal distribution centered around a complex, nonlinear function of the genes:

Assume:

$$Y_i \sim Normal(f(X_i), \sigma^2) \quad (20)$$

where $f(X_i)$ captures how genes and gene pairs influence the corrected phenotype. We then break $f(X_i)$ into a sum of k smaller functions where each of these functions captures the effects from a smaller subset

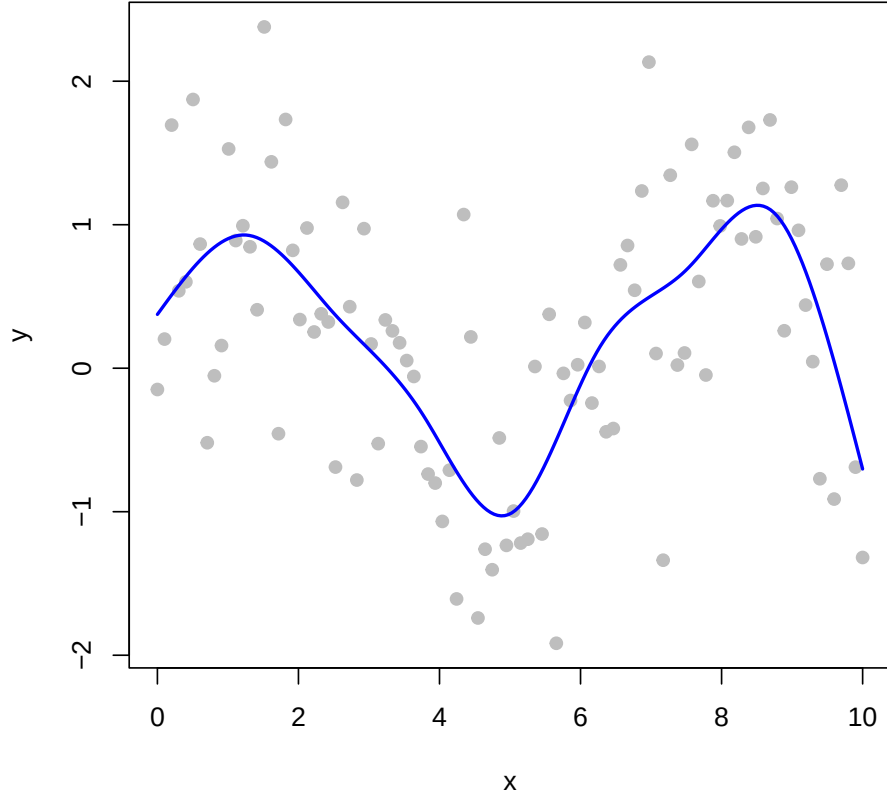


Figure 10: **Fitting with Natural Splines**

An illustration of natural spline smoothing applied to noisy, non-linear data. The gray points represent raw observations, while the blue curve shows the flexible fit obtained using natural spline basis functions. This demonstrates how splines can capture complex, multi-modal relationships without requiring a parametric form. As the degrees of freedom (df) in $ns(\cdot)$ increase, the fitted line becomes more flexible and approaches interpolation of the data, at the cost of smoothness.

of genes and gene pairs:

$$f(X_i) = \sum_{h=1}^k f^{(h)}(X_i) \quad (21)$$

$$f^{(h)}(X_i) = \sum_{g_1=1}^p \tilde{X}_{ig_1} \beta_{g_1}^{(h)} + \sum_{g_1=2}^p \sum_{g_2 \leq g_1} \tilde{X}_{ig_1 g_2} \beta_{g_1 g_2}^{(h)} \quad (22)$$

where the main effects are captured by $\sum_{g_1=1}^p \tilde{X}_{ig_1} \beta_{g_1}^{(h)}$ and the pairwise interactions are captured by $\sum_{g_1=2}^p \sum_{g_2 \leq g_1} \tilde{X}_{ig_1 g_2} \beta_{g_1 g_2}^{(h)}$.

We use a sparsity-inducing spike and slab prior over each subset $S \subset \{1, 2, \dots, p\}$:

$$P(\beta_S^{(h)} \mid \eta) = (1 - \prod_{g \in s} \eta_{gh})\delta_0 + (\prod_{g \in s} \eta_{gh})\psi(\beta_S^{(h)}) \quad (23)$$

where the spike, δ_0 , is a point mass at zero

$$\delta_0 = P(\beta_S^{(h)} = 0) > 0 \quad (24)$$

and the slab, $\psi(\beta_S^{(h)})$, is a multivariate normal

$$\psi(\beta_S^{(h)}) = \text{Normal}(\beta_S^{(h)} \mid \mathbf{0}, \sigma^2 \sigma_\beta^2 I) \quad (25)$$

where σ_β^2 is estimated via Empirical Bayes.

Let $\eta = \{\eta_{gh}\}$ be a binary matrix of inclusion indicators, where $\eta_{gh} = 1$ if gene g is included in function h , and 0 otherwise. Define the active gene set for function h as:

$$A_h = \{g : \eta_{gh} = 1\}. \quad (26)$$

Then, the prior on η_{gh} is given by:

$$P(\eta_{gh}) = \tau_h^{\eta_{gh}} (1 - \tau_h)^{1 - \eta_{gh}} \cdot I(A_h \not\subseteq A_m \ \forall m \text{ or } A_h = \emptyset), \quad (27)$$

where τ_h is the prior probability of a gene or gene-pair being included in function h . The inclusion probability follows a Beta prior:

$$\tau_h \sim \text{Beta}(M, \gamma), \quad (28)$$

with M and γ fixed to control sparsity. We set $\gamma = p$, the total number of genes considered.

This modeling framework allows us to prioritize genes and gene-pairs that are contributors to IA in DAISY children while naturally handling both non-linear effects and higher-order interactions. The use of natural splines introduces flexibility, while the spike-and-slab prior ensures sparsity and interpretability—only genes and interactions with substantial evidence are retained.

Importantly, this framework is hierarchical: after identifying informative chromosomes and chromosomal interactions, we can restrict the gene-level interaction model to only relevant chromosomal pairs. This two-stage approach melds the need for computational feasibility with a natural biological partition, enabling genome-scale modeling without exhaustively evaluating all gene-gene combinations.

The resulting posterior inclusion probabilities (PIPs) serve as importance scores. This procedure

offers a discovery-oriented path for understanding the polygenic, interactive architecture of IA risk in DAISY children, allowing for focused downstream analysis of variant-level contributions within prioritized gene-gene interactions.

2.5 Prioritized Gene-Pair SNP Synergies

To further investigate the fine-scale genetic architecture of gene-gene interactions, we construct a synergy matrix $\mathbf{S}$ for each gene pair g_1, g_2 identified in Section 2.4.2 as interacting. Similar to Section 2.3.3, each matrix $\mathbf{S}$ encodes the pairwise synergy between SNPs i and j , where $i \neq j$ and both i and j belong to the union of SNPs mapped to genes $g_1 \in \mathcal{G}_1$ and $g_2 \in \mathcal{G}_2$. That is, $i, j \in \mathcal{S}_1 \cup \mathcal{S}_2 = \mathcal{S}_{sub}$, where $\mathcal{S}_1, \mathcal{S}_2 \subset \mathcal{S}$ and $\mathcal{S}_1 \mapsto \mathcal{G}_1, \mathcal{S}_2 \mapsto \mathcal{G}_2$, with $\mathcal{G}_1, \mathcal{G}_2 \subset \mathcal{G}$.

Each entry S_{ij} quantifies the pairwise epistatic interaction strength—captured as synergy—between SNPs i and j with respect to the disease phenotype. This enables the identification of specific variant combinations that may underlie the observed gene-level interaction signal.

To visualize and further explore these interactions, we construct a graph $G = (V, E)$ for each gene pair, where nodes V represent SNPs and edge weights E are determined by the corresponding synergy scores S_{ij} . A central challenge in this framework is deciding which edges to include in the graph: while statistical approaches such as permutation testing could be used to assess significance, such methods would require thousands of permutations per SNP pair and are therefore computationally intractable for discovery even within gene-pairs.

As a pragmatic alternative, we define edge inclusion based on quantile thresholds of the synergy distribution. Specifically, we include edges whose synergy magnitude exceeds the $1 - \alpha$ quantile, thereby prioritizing SNP pairs with the strongest epistatic signals—whether positive or negative. We set $\alpha = 0.1$ to focus on the top 10% most extreme interactions. While this approach does not explicitly model statistical significance, it enables interpretable and scalable exploration of variant-level interactions, and lays the foundation for future work incorporating rigorous uncertainty quantification.

3 Results

3.1 Overview

3.2 Plots

3.3 Limitations

4 Discussion

5 Conclusion

6 References

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A Data Preprocessing

B Code

C Additional Figures & Tables